

Problem 3.11

You invest \$1,000 now, at an annual simple interest rate of 6%. What is the effective rate of interest in the fifth year of your investment?

Solution

For simple interest,

$$a(t) = 1 + it,$$

with

$$i = 6\% = 0.06.$$

Also,

$$a(t) = \frac{A(t)}{A(0)}.$$

Since

$$A(0) = 1000,$$

we compute

$$A(5) = (1 + 0.06 \cdot 5)(1000) = (1.30)(1000) = 1300,$$

and

$$A(4) = (1 + 0.06 \cdot 4)(1000) = (1.24)(1000) = 1240.$$

Now, the effective rate in the fifth year is

$$i_5 = \frac{A(5) - A(4)}{A(4)}.$$

Substituting,

$$i_5 = \frac{1300 - 1240}{1240}.$$

Therefore,

$$i_5 = 0.04838 \approx 4.84\%.$$